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Section: H
Course: Software Engineering Lab 4 - VibeCoding
Scenario 18: Blackjack vs Dealer
LLM used: Claude (claude.ai)

Chat History

Prompt 1

Here are cards.py, game.py and main.py from the Scenario 18 blackjack repo. Before changing anything, walk through hand_value by hand for A+9, A+9+5 and A+A+9 and tell me what the current code returns and what it should return.

Response 1

The assistant traced the generator expression in hand_value. Every Ace is added as 11 and never demoted, so A+9 returns 20 (correct by luck), A+9+5 returns 25 instead of 15, and A+A+9 returns 31 instead of 21. Any hand holding an Ace therefore busts far too early, and the bug is in cards.py only. I reproduced this by running the game with a seeded deck: the hand AC 8C 9S was scored 28 and reported as a bust, when the true total is 18. That run is the before video.

Prompt 2

Rewrite hand_value so an Ace counts as 11 whenever that does not bust the hand and as 1 when it must. Keep it in cards.py, keep it standard library only, and also give me is_blackjack and is_soft helpers.

Response 2

hand_value now accumulates every Ace as 11 while counting them, then subtracts 10 per Ace in a loop for as long as the total is over 21. is_blackjack checks for exactly two cards totalling 21, and is_soft compares the best total against the all-Aces-as-1 total. I checked A+9=20, A+9+5=15, A+A+9=21, A+A+A+8=21, A+K=21 and K+Q+2=22, and the three card hand 10+6+A=17.

Prompt 3

Now fix the round in game.py. It has to handle natural blackjack at 3:2, player bust, dealer bust, the dealer drawing to 17, and pushes. Chips must persist across rounds with a wager the player chooses, and a round must settle the bankroll exactly once. No files, no database, keep everything in memory.

Response 3

The round now asks for a wager first, deals, and resolves naturals before any player action so no hit is possible on a finished round. All bankroll changes go through one settle method that takes

a result of blackjack, win, lose or push, which is why a round can only settle once. The player loop exits on a stand, a bust or a quit, and the dealer then draws while under 17. A push returns the wager. I asked why settle was separated out and the reason given was that the earlier code adjusted chips in three different branches, which is how a round could be paid twice.

Prompt 4

Add feedback for the actual cards that are drawn and for the outcome of the round, and make sure an unknown command or an invalid wager cannot change the bankroll. Also handle the deck running out. Then list the cases I should test.

Response 4

Every draw now prints the card, the dealer's hole card is announced when it is revealed, and the chip count is printed after each settlement. ask_wager rejects non numeric input, zero or negative amounts and amounts above the bankroll, and re-asks without touching the chips. An unrecognised command prints a hint and re-prompts. Both draw sites check for an empty deck and cancel the round with the wager returned. Tested: one and multiple Aces, natural blackjack for the player, for the dealer and for both, player bust, dealer bust, dealer drawing to 17, push, valid and invalid wagers, repeated and unknown commands, a deck forced empty, and quitting at the wager prompt and mid round. The bankroll was 100 chips before and after every invalid input run.

Outcome

The four tasks in the repository README were completed in four prompts. The before video shows the hand AC 8C 9S scored as 28 and declared a bust. The after video shows the same seeded hand scored as 18, together with the wager prompt, the soft hand marker, the dealer drawing to 17 and the push being settled.